

Unit 3 — Compactified Scale Duality: a geometric route toward the unification of quantum theory and gravitation

One scale field, from black holes to inflation, and a bridge to intermittent quantum
existence

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Abstract

The unification of quantum mechanics and general relativity remains open. We propose that their two faces emerge from a single structure: the compactification of the scale axes, spatial and temporal, into circles — $\mathcal{M} = M_{3+1} \times S_\psi^1 \times S_\chi^1$ under $\mathbb{Z}_2 \times \mathbb{Z}_2$. A single scale field, coupled non-minimally via $f = 1 + \beta(\cos \psi + \cos \chi)$ and carrying a bounded potential $V = \alpha(2 - \cos \psi - \cos \chi)$, governs both regimes. We establish: (i) natural inflation at locked coupling, predictive — (n_s, r) and a constraint on β ; (ii) the absence of any regular black hole in the *static* sector (closure), robust to un-freezing the temporal axis; (iii) in collapse, that the verdict on whether curvature is capped is *reduction-dependent* and open — it hinges on the nature of the pole ∞^* (§V); (iv) a bridge linking the temporal axis χ to the quantum time of the Quantum Existence Discontinuity hypothesis (HDEQ), gravity modulating time. The field-space metric is positive-definite ($f > 0$, B1), the viable band the open interval $\beta \in (-0.5; -0.15)$; the scale axes are internal circles (σ -model), with no time-like dimension or ghost (F2); free propagation does not violate Lorentz. The existence factor is derived exactly, $\varepsilon = E(k)/K(k)$. α is fixed by the scalar amplitude A_s ($\alpha \approx 0.5\text{--}1.2 \times 10^{-9} M_{\text{Pl}}^4$, GUT scale), whence $\tau_0 \approx 3.5 \times 10^{-37}$ s *at the inflationary* $f_s \approx 5\text{--}6$ and a running prediction $dn_s/d \ln k \approx -6.5 \times 10^{-4}$ (A1); the rescaling $\hat{H}_{\text{eff}} = \varepsilon \hat{H}_{\text{sys}}$ is shown via a Page–Wootters relational clock (D1); the gravitational modulation $\tau_0(R)$ reduces in the laboratory to standard GR time dilation, but $\tau_0 \rightarrow \infty$ at the collapse curvature R_c (D4). **Honesty about scope.** The inflationary sector reproduces, without improving it, the periodic non-minimal natural inflation of the literature; the novelty is not there. The rigorous test (E1) shows the clock sector to be non-discriminating ($\varepsilon = 1 - 10^{-112}$): the CMB is the sole judge, and the framework’s proper signature *would be* the cross-sector correlation $\beta(\text{CMB}) = \beta(\text{closure})$ — a conjecture conditional on the reduction (§V), not a theorem. Each statement is tagged by its actual status.

Keywords: scale duality; scalar-tensor gravity; non-minimal coupling; natural inflation; regular black holes; intermittent quantum existence; intrinsic decoherence; relational clock; unification.

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Epistemic status: [proof] exact algebra/computer algebra · [numerical] reproducible integration · [conjecture] plausible, unproven · [heuristic] analogy · [speculation] open · [open] to be established · [to check] external ref. No conjecture is presented as a theorem.

I Introduction

Reconciling quantum mechanics and general relativity has been an open problem for a century. On the gravitational side, two requirements bear on any Planck-regime physics: removing singularities [3–7] and accounting for the primordial accelerated expansion [8, 9]. On the quantum side, decoherence and the selection of a preferred basis must be explained without an ad hoc external environment [2, 20]. Established programmes (strings, loop gravity, noncommutative geometry, hybrid semiclassical gravity) address these fronts by distinct routes.

We explore a single geometric hypothesis. The logarithmic scale axis, usually non-compact, is taken to be a circle: by stereographic projection, the Planck scale (pole P) and a « unified » scale (pole ∞^*) become two poles of a single manifold [1]. There are two such axes — spatial (ψ) and temporal (χ) — whence the arena $\mathcal{M} = M_{3+1} \times S_\psi^1 \times S_\chi^1$. The spatial axis governs the gravitational/cosmological face [1]; the temporal axis, the quantum face — the time of the Quantum Existence Discontinuity hypothesis (HDEQ) [2].

An ontological clarification, up front. HDEQ does not posit a discrete time. Time remains a continuous parameter; what is intermittent is the active existence of quantum systems, in cycles of duration τ_0 (active fraction ε). The qualifier « temporal » for the χ axis denotes the clock role it plays (§VI), not a metric time-like signature.

Scope and epistemic honesty. This work does not claim to solve unification on its own. It proposes a coherent and falsifiable direction: modeling decisions are assumed, results are tagged by their actual status, and it is for the community to test, extend or refute it.

II Formal framework: the arena and the action

Postulate 1 (compactification). The fundamental manifold is $\mathcal{M} = M_{3+1} \times S_\psi^1 \times S_\chi^1$, the scale axes being circles via $\psi = 2 \arctan[\ln(\lambda/\ell_P)]$ and $\chi = 2 \arctan[\ln(\tau/\tau_P)]$. The poles $\pm\pi$ are identified (∞^*); 0 is the Planck point P .

II.A Action, field-space metric (B1) and viable band

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{2} f R - \frac{1}{2} \omega ((\partial\psi)^2 + (\partial\chi)^2) - V \right], \quad f = 1 + \beta(\cos\psi + \cos\chi), \quad V = \alpha(2 - \cos\psi - \cos\chi). \quad (1)$$

[proof] (B1) In the Einstein frame, the field-space metric $\mathcal{G}_{ab}$ is positive-definite $\Leftrightarrow f > 0$: a kinetic part ω/f (positive-definite) plus a rank-1 positive-semidefinite non-minimal part. The viable band is the open interval $\beta \in (-0.5; -0.15)$ (at $\beta = -1/2$, $f \rightarrow 0$ at P). [proof] (F2) S_ψ^1, S_χ^1 are internal circles (Euclidean σ -model target), not spacetime dimensions; χ is not a time-like coordinate $\rightarrow$ no ghost, no closed time-like curves. Winding = the topological scale charge.

III The gravitational and cosmological sector (« infinitely large »)

[proof] At frozen fields, Einstein gravity emerges; at the symmetric vacuum P ($V = 0$), the cosmological constant vanishes. At the pole ∞^* , the source reduces to a pure vacuum energy, giving a locally de Sitter space of rate $H^2 = 2\alpha/[3(1 - \beta)]$ (one axis) or $4\alpha/[3(1 - 2\beta)]$ (two axes), with finite Kretschmann scalar $K = 24H^4$.

In homogeneous cosmology, the canonical field $\varphi = f_s \psi$ sees the potential $V = \alpha[1 - \cos(\varphi/f_s)]$: a natural inflation at locked non-minimal coupling — $f = 1 + \beta \cos\psi$ has the same structure as the potential (studied independently in [16]). [numerical] Integration starts at ∞^* (de Sitter phase) and rolls down to P ($\Lambda = 0$), with graceful exit. Slow-roll predictions (Einstein frame):

β	$f_s (M_{\text{Pl}})$	n_s	r	status
0	7	0.960	0.078	r too large
+0.3	7	0.949	0.128	excluded
-0.3	5	0.958	0.017	compatible (edge)
-0.3	6	0.959	0.027	compatible (edge)
-0.3	7	0.960	0.036	at the BK18 edge

Table 1: Slow-roll (n_s, r) at $N = 55$. Target Planck+BK: $n_s = 0.9649 \pm 0.004$, $r < 0.036$.

For the natural sign $\beta \geq 0$, $r \gtrsim 0.08$ is excluded [12, 13]. The viable window is a narrow band $\beta \in (-0.5; -0.15)$ with $f_s \approx 5\text{--}6 M_{\text{Pl}}$ (open interval, cf. II.A), giving $r \approx 0.02\text{--}0.04$ correlated with $n_s \approx 0.96$. [numerical] *Honesty caveat (P1)*. Across this band $n_s \approx 0.958\text{--}0.961$, i.e. $\sim 1\text{--}2\sigma$ below the Planck central value: the framework lives *at the edge* of the contour. The CMB data provide a constraint on the scale-duality coupling β within this framework (the (n_s, r) phenomenology being shared with [16, 24, 25]), testable by the Simons Observatory [27] and LiteBIRD [14] ($\sigma(r) \sim 10^{-3}$) by 2030.¹

[numerical] (A1) Second-order slow roll with the exact field-space metric reproduces Table 1 to better than 0.5%, and fixes α via $A_s = 2.1 \times 10^{-9}$: $\alpha \approx 0.5\text{--}1.2 \times 10^{-9} M_{\text{Pl}}^4$, an inflationary scale $V^{1/4} \sim 1.1\text{--}1.4 \times 10^{16}$ GeV (GUT). At the inflationary $f_s \approx 5\text{--}6$ this gives $\tau_0 = 2\pi\sqrt{\omega/\alpha} \approx 3.4\text{--}3.8 \times 10^{-37}$ s (self-consistent: the clock period now uses the *same* f_s as inflation, resolving the earlier $f_s = 57$ mismatch which gave $r \approx 0.14$, excluded). New prediction, stable across the band: the running $dn_s/d\ln k \approx -6.5 \times 10^{-4}$ — compatible with Planck (-0.004 ± 0.007). (Remaining: mode-by-mode Mukhanov–Sasaki, expected sub-percent; (n_s, r, α) are computed on the single-field representative trajectory, the full two-field treatment — f offset at $\chi = 0$, turning effects — remaining an untested assumption, track A2.)

Honesty about novelty. This sector reproduces, without improving it, the periodic non-minimal natural inflation of Ferreira–Notari–Simeon [16], its scalar-tensor extension [24], and the $(n_s, r, \text{running})$ analyses against BICEP/Keck [25]. The novelty of the present work is therefore not the inflationary phenomenology, but the attempted cross-sector unification (§V).

IV The static sector: no regular black hole (closure)

[proof][numerical] At one field [1], the static radial profile is closed: the frozen branch $\psi \equiv -\pi$ is exactly Schwarzschild–de Sitter; the rolling branch develops a singular horizon (collapse of $B = e^{\Phi+\Lambda}$, negative Misner–Sharp mass, divergent R). We rest the closure on the *robust* argument rather than on a single out-of-band tableau: by Derrick + topology, the static profile (connecting ∞^* to P) is an *unprotected half-soliton* $N = \frac{1}{2}$, hence contractible — no regular interpolating solution exists, for any β in the band. The integer soliton $N = 1$ has mass $8\sqrt{\omega\alpha} \approx 2\text{--}5 \times 10^{15}$ GeV (GUT scale; $\approx 3 \times 10^{15}$ at the central $f_s \approx 6$, with α fixed in A1) — a value harmonized with the corrected f_s (P1).

Un-freezing the temporal axis. [proof] Making χ dynamic, $\chi \equiv 0$ remains an exact solution and reproduces the closure, but the coupling of χ to R renormalizes gravity: on the slice $\chi = 0$, $f_{\text{eff}} = (1 + \beta) + \beta \cos \psi$, and the stability threshold shifts ($\beta_c = -1/4$ for two axes). The closure conclusion is robust; the precise threshold value is reduction-dependent ($-1/3$ mono-axis, $-1/4$ on the $\chi = 0$ slice, $-1/6$ on the diagonal — see §V).

V The pole ∞^* : a structural tension between closure and unwinding

[numerical] Does *dynamical* (non-static) collapse cap the curvature? The answer turns on the nature of the pole ∞^* — which is *not* reduction-independent. This is the crux, and we record it as the central structural result of this synthesis: an open tension, not a settled verdict.

¹The CMB-S4 project (DOE–NSF), long the reference instrument for this test, was halted: on 9 July 2025 the DOE and NSF stated they could no longer support the project (orderly shutdown), which is therefore no longer funded [15]. The falsifiability stands, carried by the Simons Observatory (Chile, operating), LiteBIRD (~ 2032) and BICEP/Keck (ongoing). [to check]

Mono-axis / Jordan reading. On the single active axis (χ frozen), for the Jordan potential V , ∞^* is a maximum ($V''(-\pi) = -\alpha < 0$). Homogeneous spherical collapse with the field rolling then drives ψ *through* ∞^* : the curvature is not capped, no regular core forms. In this reading the two verdicts — graceful exit in cosmology, no regular core in collapse — share the cause « ∞^* is a hilltop ».

Two-field / Einstein-frame reading. But the cosmology that fixes (n_s, r) is, strictly, a two-field system on the diagonal $\psi = \chi$. [numerical] In the Einstein-frame diagonal landscape $U = V/f^2$, the nature of ∞^* changes sign at $\beta = -1/6$: for $\beta > -1/6$ it is a maximum (rolls), but for $\beta < -1/6$ — i.e. across the *entire viable band* $\beta \in (-0.5; -0.15)$ — ∞^* is a *trapped local minimum* (a *false vacuum*) behind an internal barrier θ_b . In this reading a collapse driving the field to ∞^* could be *retained* there (de Sitter core) rather than rolling through — which would instead point toward a capped curvature, i.e. a possible regular core.

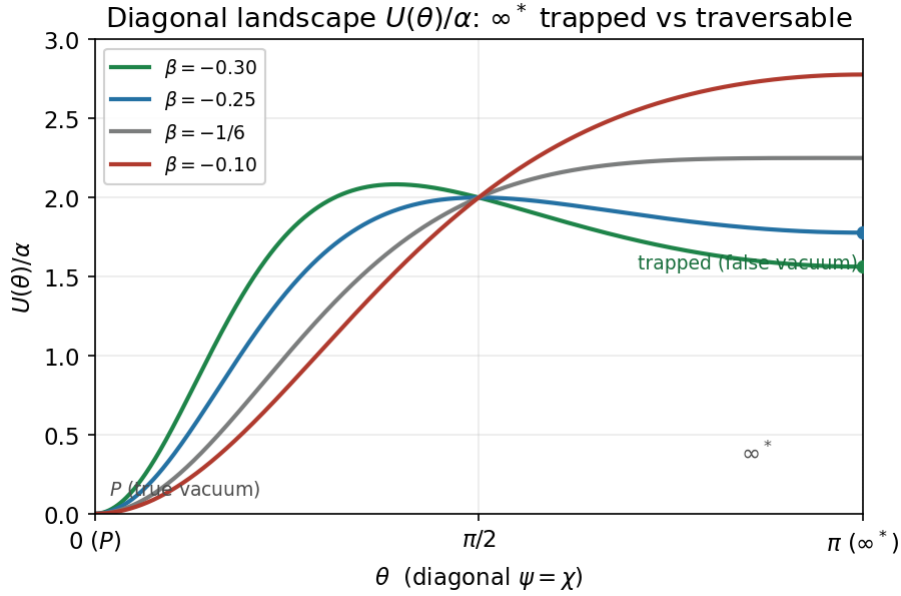


Figure 1: [numerical] Einstein-frame diagonal landscape $U(\theta)/\alpha = 2(1 - \cos \theta)/(1 + 2\beta \cos \theta)^2$ ($\psi = \chi = \theta$, exact closed form). For $\beta < -1/6$ (green, blue — the entire viable band) ∞^* ($\theta = \pi$) is a *trapped local minimum* behind a barrier (false vacuum); for $\beta \geq -1/6$ (grey, red) it is a maximum (the field rolls through). The closure wants ∞^* trapped, the unwinding wants it traversable: the same pole cannot be both.

The tension. The two readings disagree on the very nature of ∞^* , hence on whether collapse is regularized. They are not cosmetic. The *static* closure rests on ∞^* being a trap (Derrick: the field cannot connect ∞^* to P , so the only static solution is the frozen Schwarzschild–de Sitter), while the literal « unwinding » picture (the parchment, the field winding *through* ∞^*) requires ∞^* to be traversable. The same pole cannot be both, and the band $\beta < -1/6$ that delivers the closure is exactly the band that obstructs dynamical unwinding. We therefore record:

- the **static** closure (no regular black hole) is robust — it follows from Derrick + topology (contractible half-soliton $N = \frac{1}{2}$) and holds across reductions. [proof]
- the **dynamical** collapse verdict is reduction-dependent and *open*: rolls-through-and-not-capped in the mono-axis/ V reading, but a trapped false vacuum (hence a candidate for a capped, regular core) in the two-field Einstein-frame reading across the viable band. [conjecture][open]

A failed shortcut. One might hope that genuine multi-field dynamics (a « spiral »: the field winding off the diagonal) suppresses r or sustains the unwinding. [numerical] A full two-field integration with an injected lateral boost shows it does not: the turning is transient ($\omega/H \approx 1.6$ initially) and damped by Hubble friction well before the CMB window; r stays ≈ 0.046 , never suppressed. The reason is structural — P is a point well, not a ring valley, so nothing sustains an orbit. The « unwinding » is therefore best read as a *topological/spatial* statement (the integer soliton charge N), and the literal *temporal* unwinding (a wound $N \geq 1$ trajectory through ∞^*) as a *conditional* signature (log-periodic features) available only *outside* the viable band — not as an established dynamical mechanism.

The one quantum-sector opening. If a regularization exists in the viable band, the candidate is not classical roll-through but the quantum clock: at the collapse curvature $R_c = 2\alpha/|\beta|$, the effective mass of χ vanishes and $\tau_0 \rightarrow \infty$ (D4, §VI) — the cycle no longer closes, which could forbid the classical roll-through and retain the field at ∞^* . This is the only avenue by which the unification loop might close. [speculation]

VI The quantum sector and the bridge (« infinitely small »)

VI.A The existence factor exactly derived (D1)

[proof] The oscillation of χ in V is a pendulum (sine-Gordon). For an amplitude $\chi_{\max}$ (elliptic modulus $k = \sin(\chi_{\max}/2)$), the identity for $\langle \sin^2 \rangle$ gives the exact form (verified by integrating $\chi'' = -\sin \chi$ to 10^{-16}):

$$\varepsilon = \langle \cos^2(\chi/2) \rangle = \frac{E(k)}{K(k)}, \quad \tau_0(k) = 4\sqrt{\omega/\alpha} K(k).$$

At small amplitude $\varepsilon \rightarrow 1 - \chi_{\max}^2/8$ and $\tau_0 \rightarrow 2\pi\sqrt{\omega/\alpha}$. Approaching the separatrix (∞^*), $\varepsilon \rightarrow 0$ (suspension). ε is therefore no longer a free parameter: it is $E(k)/K(k)$, fixed by the amplitude.

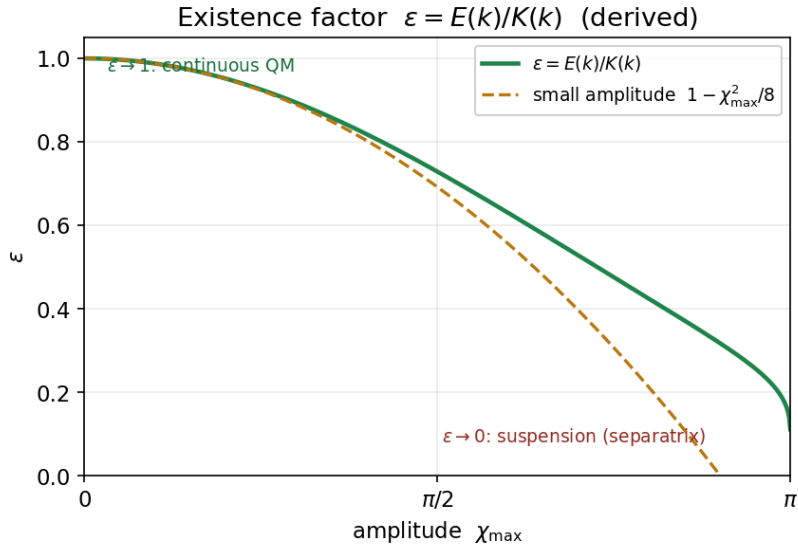


Figure 2: [numerical] The existence factor $\varepsilon = E(k)/K(k)$, $k = \sin(\chi_{\max}/2)$ (verified against the pendulum $\langle \cos^2(\chi/2) \rangle$ to 10^{-6}). At small amplitude $\varepsilon \rightarrow 1 - \chi_{\max}^2/8$ (continuous QM); at the separatrix $\varepsilon \rightarrow 0$ (suspension). ε is derived, not free.

VI.B Operatorial realization (D1) and gravitational modulation (D4)

[conjecture] (realized, D1) χ is a Page–Wootters relational clock [22]: on $\mathcal{H}_{\text{clock}} \otimes \mathcal{H}_{\text{sys}}$ with a timeless constraint, the system evolves with respect to χ — defining « temporal » as a clock role, with no time-like signature (answer to F2). For *constant* $\hat{H}_{\text{sys}}$, the evolution gated by $g(\chi) = \cos^2(\chi/2)$ gives over one period $\hat{U}(\tau_0) = \exp(-i\hat{H}_{\text{sys}}\varepsilon\tau_0)$, i.e. $\hat{H}_{\text{eff}} = \varepsilon\hat{H}_{\text{sys}}$ — the Floquet result $\Omega_F = \varepsilon\Omega_R$ of [2], here verified for constant $\hat{H}_{\text{sys}}$ (gated qubit, error $\sim 10^{-11}$); the non-commuting case (full Floquet) is estimated negligible but *not* proven. [open] derive the timeless constraint from the χ action; exactness of the two-state reduction.

[proof] (D4) The coupling of χ to R shifts its effective mass: $\tau_0(R) = 2\pi\sqrt{\omega/(\alpha + \frac{1}{2}\beta M_{\text{Pl}}^2 R)}$. Two effects, not to be confused. (A) $\sqrt{-g_{00}} \approx 1 + \phi/c^2$: standard gravitational dilation (τ_0 is a proper-time scalar), observable but not new — it is GR. (B) the dynamical shift $\delta\tau_0/\tau_0 \approx -(\beta/4)(M_{\text{Pl}}^2 R/\alpha) \propto \text{local } R$: it vanishes in vacuum and is $\sim 10^{-81}$ in matter — unobservable. But at $R_c = 2\alpha/|\beta|$ (the collapse threshold of §V), the effective mass of χ vanishes: $\tau_0 \rightarrow \infty$, the quantum clock freezes. The same R_c that triggers the descent makes the cycle diverge — a concrete realization of trapping at ∞^* [conjecture]. ($\tau_0(R)$ is small-amplitude; the exact behavior near R_c requires the full anharmonic potential.)

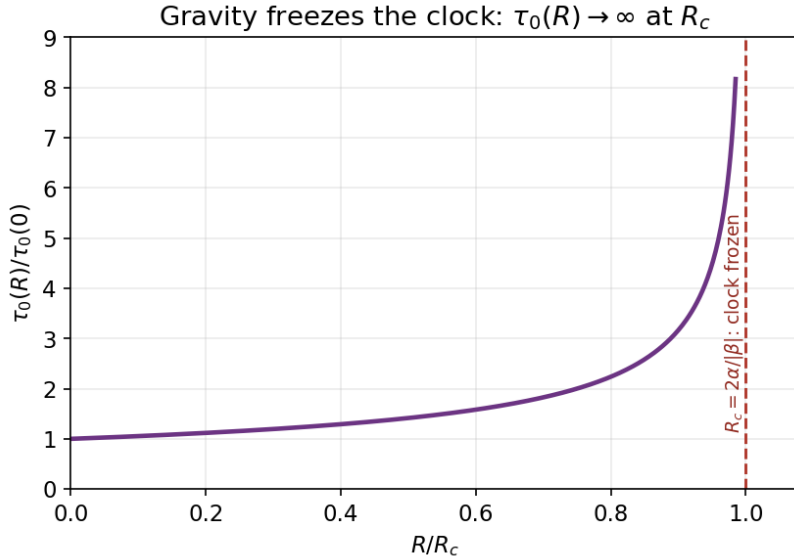


Figure 3: [numerical] Gravitational modulation $\tau_0(R)/\tau_0(0) = 1/\sqrt{1 - R/R_c}$, $R_c = 2\alpha/|\beta|$. The clock period diverges as the curvature approaches the collapse threshold R_c : the quantum clock freezes — the one candidate route for the quantum sector to regularize collapse.

VI.C Lorentz safety of the quantum time

[proof] $\tau_0 \sim 10^{-37}$ s corresponds to $\hbar/\tau_0 \approx 10^{13}$ GeV; one must check it induces no Lorentz violation (photon time-of-flight, LHAASO). An intermittent existence violates Lorentz only through a momentum-dependent dispersion modification. Now (i) the proper evolution of $\hat{H}$ is exactly phase-linear to all orders in τ_0 (Proposition 1 of HDEQ); (ii) the Maxwell action is Weyl-invariant in $D = 4$, so the photon is blind to the conformal factor $f(\chi)$. Free-photon dispersion remains exactly $\omega = |p|c$: no LHAASO signal. The only correction (QED trace anomaly) is logarithmic, not power-law in E/E_{QG} — no linear LIV term. [to check]

VI.D Cross-sector test (E1): the CMB is the sole judge

[numerical] On the α fixed in A1, $\tau_0 \in [2.5; 5.3] \times 10^{-37}$ s. Decisive point: unification removes the freedom in ε . The bridge ties ε to the present field displacement, which equals dark energy;

numerically $\psi_{\text{today}} \sim 10^{-56}$ rad, so $\varepsilon = 1 - 3 \times 10^{-112}$. Consequences: the Yb⁺ effect $\Delta\Gamma/\Gamma = 1 - \varepsilon^2 \approx 10^{-112}$, intrinsic decoherence below the floor, the resonance at $\Delta E \approx 1.2 \times 10^{13}$ GeV — the entire HDEQ clock sector is unobservable. *Caveat (open)*. The framework sets $\Lambda = 0$ at the vacuum P , whereas observed dark energy is non-zero; the identity « present residual = dark energy » that fixes $\varepsilon = 1 - 10^{-112}$ is not derived, and the reconciliation with the observed Λ remains open (a residual-quintessence reading, ε slightly below 1, would carry $\Lambda > 0$ at the cost of fine-tuning). The « clocks non-discriminating » conclusion holds in both readings. [open]

Verdict E1 (rigorous). The framework passes the consistency test, but the confrontation is not symmetric: the clock side is satisfied trivially ($\varepsilon \approx 1$). All the falsifiability burden therefore rests on the CMB ($n_s \approx 0.96$, $r \approx 0.02\text{--}0.04$, running $\approx -6.5 \times 10^{-4}$). The « same (α, ω) for the CMB and the clocks » is met automatically on the clock side — it should *not* be presented as a discriminating test.

VII Overview and falsifiability criterion

The vision is an ansatz, not a result: a single arena whose known laws are its two limiting regimes. (*Infinitely large:*) slow fields $\rightarrow$ general relativity, $\Lambda = 0$ at vacuum, inflation. (*Infinitely small:*) circle dynamics $\chi \rightarrow$ intermittent existence, decoherence (HDEQ). The hinge is the χ axis, which couples to curvature and modulates quantum time via gravity. The two faces are joined by a bridge now partially proven (ε derived, gating included; the full operator construction still open).

The truth criterion is numerical, but asymmetric: the CMB (n_s, r , running) is the judge; the atomic clocks provide a consistency constraint automatically met ($\varepsilon \approx 1$), not a discriminating test. The signature *proposed* for this framework — which no isolated cosmological observable possesses, since $(n_s, r, \text{running})$ are shared with [16, 24, 25] — is the cross-sector correlation $\beta(\text{CMB}) = \beta(\text{closure})$: the same coupling should fit inflation and satisfy the black-hole closure thresholds. *This correlation is a conjecture conditional on the reduction* (§V), not a theorem: with three candidate thresholds ($-1/3, -1/4, -1/6$) « $\beta(\text{closure})$ » is not yet a single number, so settling the reduction (track A2) is the prerequisite to turning this into a genuine discriminator. That is where falsifiability would live, testable via LiteBIRD / Simons Observatory.

VIII Epistemic status and limitations

Content	Status
Finite action; GR limit; natural inflation at locked coupling, (n_s, r) and a constraint on β ; robust <i>static</i> closure (Derrick + topology); $\mathcal{G}_{ab}$ positive-definite (B1); σ -model health, no ghost/CTC (F2); photon Lorentz safety; $\varepsilon = E(k)/K(k)$ derived; α fixed by A_s , running $\approx -6.5 \times 10^{-4}$ (A1); self-consistent $f_s \approx 5\text{--}6$ and τ_0 .	[proof][numerical]
$\hat{H}_{\text{eff}} = \varepsilon \hat{H}_{\text{sys}}$ via a Page–Wootters clock, restricted to constant $\hat{H}_{\text{sys}}$ (D1); $\tau_0(R)$ derived, $\tau_0 \rightarrow \infty$ at R_c (D4); E1: clock sector non-discriminating, CMB sole judge.	[conjecture] (realized)
Central open tension: the nature of ∞^* (traversable vs trapped) governs the <i>dynamical</i> collapse verdict and the cross-sector signature; reduction not settled (track A2). Derive the Page–Wootters constraint from the χ action; reconcile $\Lambda = 0$ vs observed dark energy; inhomogeneous collapse; quantum trapping at ∞^* (regularization); Born rule; single-branch selection; mode-by-mode Mukhanov–Sasaki.	[open][speculation]

The distinction of statuses is not an ornament but a constitutive part of the method: it ensures this work can be reused without any statement exceeding what is actually established.

IX Conclusion

A single scale-duality field, on an arena with compactified axes (internal circles, not spacetime dimensions), generates a predictive and falsifiable inflationary cosmology, settles the *static* regular-black-hole question by one structural cause (Derrick + topology), and offers a bridge toward the quantum time of HDEQ now partially proven: the existence factor $\varepsilon = E(k)/K(k)$ is derived, the active/inactive mechanism and the Lorentz-safe free propagation established. The *dynamical* regularization, however, is not settled: it hinges on the nature of the pole ∞^* , traversable in one reduction and a trapped false vacuum in another — a tension we record as the central open problem, with the quantum freezing at R_c as the one candidate route to close the loop. This is not a closure of unification; it is a direction whose very nature is to be refutable. That is the price — the honest tagging of each statement — at which an individual intuition can become a collective direction.

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Reproducibility

The numerical results and symbolic verifications are reproduced by two standalone scripts (NumPy/SymPy/SciPy) with sections coded so as to be specific to this document (no collision with other script sets):

- [S1] existence factor $\varepsilon = E(k)/K(k)$ verified against the pendulum $\langle \cos^2(\chi/2) \rangle$; small-amplitude $\varepsilon \rightarrow 1 - \chi_{\max}^2/8$; $\tau_0(k) = 4\sqrt{\omega/\alpha} K(k)$.
- [S2] nature of ∞^* : diagonal threshold $\beta = -1/6$ (trapped/rolls) and transverse isocurvature mass (sign change at $\beta = -1/4$).
- [S3] inflation on the viable band $f_s = 5-6$: (n_s, r) diagonal vs mono (agreement); α , $V^{1/4}$, $\tau_0 \approx 3.5 \times 10^{-37}$ s, soliton mass (P1).
- [S4] gravitational modulation $\tau_0(R)$; $R_c = 2\alpha/|\beta|$; $\delta\tau_0/\tau_0 = -(\beta/4)M_{\text{Pl}}^2 R/\alpha$.
- [S5] spiral test (two-field integration): the turning is damped, r never suppressed.

Files: `U3_SYNTH_reproducible.py` (calculations) and `Synthese_figures.py` (Fig. 2, 1, 3). *Convention*: $M_{\text{Pl}} = 1$; α cancels in the (n_s, r) ratios. The prefix `U3_SYNTH_` and the `[S]` codes are specific to this document.

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